Rafael Garcia-Dias

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PROFESSIONAL SUMMARY

Senior AI Engineer with a PhD in Machine Learning and 10+ years of production Python experience. Currently building FLIP — the open-source federated learning platform that connects 11 NHS Trusts and serves AI development across a population of 10M+ patients. I work end-to-end: foundation-model research (multimodal vision-language models for radiology), platform engineering (FastAPI, Docker Swarm, AWS, OpenTofu) and federated MLOps (NVIDIA FLARE, Flower).

PROFESSIONAL EXPERIENCE

Senior AI Engineer — Foundation Models for Healthcare — Nov 2024 – Present

AI Centre for Value-Based Healthcare / King's College London • London, UK

Building production AI infrastructure and foundation models across an 11-NHS-Trust federation in collaboration with universities (KCL, GSTT) and industry partners (NVIDIA, deepc, Flower Labs, OneLondon).

- Core contributor to the multi-Trust platform launched publicly in Feb 2026, enabling privacy-preserving AI training across NHS firewalls without moving patient data. Microservice architecture: central hub API, per-Trust APIs, imaging and data-access services, FastAPI throughout, deployed via Docker Swarm and AWS.
- Integrated NVIDIA FLARE and Flower Framework as parallel federated-learning backends, allowing research teams to choose between two industry-standard frameworks against the same data plane.
- Designed the trust-authentication and internal-service-key model for secure outbound-only Trust-to-Hub communication via CloudFront/ALB; contributed to the OpenTofu/Terraform AWS deployment for production environments.
- Integrated DICOM imaging infrastructure (Orthanc, XNAT) and OMOP CDM structured-data layer, giving researchers unified multimodal access across imaging and EHR within each Trust's secure enclave.
- Leading development of vision-language foundation models that fuse medical imaging (X-ray, CT, MRI) with radiology reports, supporting downstream applications: automated report generation, image synthesis from reports, scan/report discrepancy detection, and LLM-based report classification.
- Validated robustness of latent representations (including LLM embeddings) so smaller task-specific models built on top remain reliable — a prerequisite for clinical translation.
- Active open-source contributor to MONAI Core and MONAI Generative, advancing foundational algorithms used by the wider medical-imaging research community.
- Drives methodology for the responsible scaling of LLMs in clinical settings — bridging cutting-edge AI capability with NHS governance, safety and real-world clinical workflows.

Machine Learning Engineer — May 2023 – Nov 2024

Floe Oral Care • London, UK

Founding ML engineer; took the product from research concept to deployed clinical tool.

- Owned the science: designed the clinical trials that validated the diagnostic concept and produced the training data for the production models.
- Shipped multiple ML models: logistic regression mapping protein concentrations to health risk, tree-based models over questionnaire data, and an OpenAI-backed RAG recommendation system.
- Built the full cloud stack on AWS with AWS CDK, including databases, ML asset storage and web infrastructure; CI/CD throughout.
- Created the Django backend, integrated the contractor-built front end and directed the front-end engineering team.
- Iterated production models post-launch based on real user data, applying SOLID principles and a strong testing culture.

Decision Scientist (Credit Risk ML) — Aug 2022 – Feb 2023 · Fixed-term, 6 months

Monzo Bank • London, UK

Improved creditworthiness modelling on a £259M lending portfolio serving 7M+ customers.

- Shipped a GBM-based overdraft-application model delivering 20–100% Gini gains over the previous baseline, deployed in Docker on GCP.
- Refactored the internal modelling library to standardise monitoring across Monzo's lending products, surfacing performance gaps across customer subgroups that had previously been invisible.
- Stack: Python, scikit-learn, BigQuery, dbt, Looker, GCP.

Research Associate — Founding Engineer, Neurofind — Aug 2018 – Aug 2022 · 4 years, fixed-term renewed 4×

Neurofind / King's College London • London, UK

- Built the backend, cloud infrastructure and ML models for Neurofind.ai, a tool aiding diagnosis of mental disorders from 3D brain MRI.
- Authored Neuroharmony, the first published out-of-sample harmonisation tool for neuroimaging — a key step toward clinical translation of imaging ML by mitigating cross-scanner bias.
- Won a £109,000 MRC grant as named investigator.
- Authored one chapter and co-authored five more in Machine Learning: Methods and Applications to Brain Disorders (Elsevier).
- Stack: Python, PyTorch, TensorFlow, scikit-learn, AWS, Docker, MLflow, Plotly, R.

PhD Fellow — Machine Learning Applied to Astrophysics — Aug 2015 – Aug 2018

Instituto de Astrofísica de Canarias (IAC) / Universidad de La Laguna • Tenerife, Spain

Applied unsupervised and supervised ML to large-scale stellar spectra surveys; developed clustering methodology that became the basis for a later book chapter and journal paper. Foundation in rigorous, reproducible scientific computing in Python (NumPy, pandas, scikit-learn, Matplotlib) on Linux.

EDUCATION

PhD, Machine Learning Applied to Astrophysics — 2018

Instituto de Astrofísica de Canarias / Universidad de La Laguna

LANGUAGES

English • Spanish • Portuguese